

Homework 1

46-921, Fall 2021

Due Wednesday, September 8 at 9:30 PM EDT

Please submit a pdf file with your responses to the questions.

Question 1:

Suppose that $X_1, X_2, \dots, X_n$ are iid with the $\text{Gamma}(\alpha, \beta)$ distribution. Determine the method of moments estimators for α and β .

Comment: This is an important case because maximum likelihood does not admit a closed form for the estimators for α and β .

Question 2:

Assume that $X_1, X_2, \dots, X_n$ are iid with mean μ and variance σ^2 . Both μ and σ^2 are unknown. Assume that the sample mean is $\bar{x} = 14.3$ and the sample standard deviation is $s = 4.2$.

- Without any further information, can you construct a valid 95% confidence interval for μ if $n = 40$? If so, do it.
- Without any further information, can you construct a valid 95% confidence interval for μ if $n = 10$? If so, do it.
- If I told you that the X_i are normal, can you construct a valid 95% confidence interval for μ when $n = 10$? If so, do it.

Question 3:

During lecture we compared three different estimators for λ when working with an iid sample from the $\text{Exponential}(\lambda)$ distribution. We concluded that, based on MSE, the “adjusted” method of moments estimator is the best choice.

Now, conduct a simulation experiment to address the following question: In the case where $n = 20$ and $\lambda = 10$, what proportion of the time does the adjusted method of moments estimator come closer to the true value of λ than does the “worst” of the three estimators (the method of moments estimator based on the second moment)? Be sure to submit your Python code and results.

Question 4:

Let $X_1, X_2, \dots, X_n$ be a iid from the following distribution:

$$f_X(x; \theta) = \begin{cases} \frac{3x^2}{\theta^3} & 0 \leq x \leq \theta \\ 0 & \text{otherwise} \end{cases}$$

Here are some facts regarding estimation of θ :

- Method of moments estimator is $\hat{\theta}_1 = \frac{4\bar{X}}{3}$
- The maximum likelihood estimator is $\hat{\theta}_2 = X_{(n)}$, where $X_{(n)}$ is the maximum of the X_i .

Here are some useful results about this distribution that will help you do this problem.

- $E(X) = \frac{3}{4}\theta$
- $V(X) = \frac{3}{80}\theta^2$
- $f_{X_{(n)}}(x) = \frac{3nx^{3n-1}}{\theta^{3n}}, \quad 0 \leq x \leq \theta$
- $V(X_{(n)}) = \frac{3n}{(3n+2)(3n+1)^2} \theta^2$

Now, do the following:

- Show that $\hat{\theta}_1$ is an unbiased estimator for θ .
- Show that $\hat{\theta}_2$ is *not* an unbiased estimator for θ , and find $\text{bias}(\hat{\theta}_2) = [E(\hat{\theta}_2) - \theta]$.
- Show that for $n > 2$, even though $\hat{\theta}_1$ is unbiased and $\hat{\theta}_2$ is not, $\text{MSE}(\hat{\theta}_2) < \text{MSE}(\hat{\theta}_1)$